

Decentralized Safe Navigation for Multi-agent Systems via Risk-aware Weighted Buffered Voronoi Cells

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Abstract

In this paper, we propose Risk-aware Weighted Buffered Voronoi tessellation, a variant of Generalized Voronoi tessellation, for decentralized multi-agent collision-free navigation. Inherited from the traditional Voronoi tessellation, a safety guarantee in terms of inter-robot collision avoidance is achieved by partitioning the joint state space of the multi-agent system into individual cells that constrain each individual agent's motion in a distributed manner. Different from many existing Voronoi tessellations-based collision avoidance approaches, our Risk-aware Weighted Buffered Voronoi Cell (Risk-aware WBVC) partition not only takes agent positional information into account, but also the motion information when determining the cell boundaries between pairwise robots. Our risk-aware WBVC relies on the novel use of Control Barrier Functions (CBF) as a measure of risk evaluation that captures to what extent the safety constraints are satisfied between pairwise robots. With that, the cell boundaries of risk-aware WBVC are determined by (1) the varying levels of relative efforts between pairwise agents to respond to potential collisions, and (2) the accumulated risk each agent experiences that is caused by the surrounding agents. This allows for an

adaptive constrained space partition among robots that balances between individual’s efforts in respecting the safety constraints and the overall threats due to other agents in the environment, e.g. an aggressive robot moving with higher speed requires a relatively larger space for responding to potential collisions, and a less-threatened robot may be expected to yield and make more room for those exposed to higher risk. Rigorous proofs of formal safety guarantees are provided and simulations are demonstrated on up to 25 robots to show the effectiveness of our method.

Keywords: Safe Control; Multi-agent System; Collision Avoidance; Robotics

1 Introduction

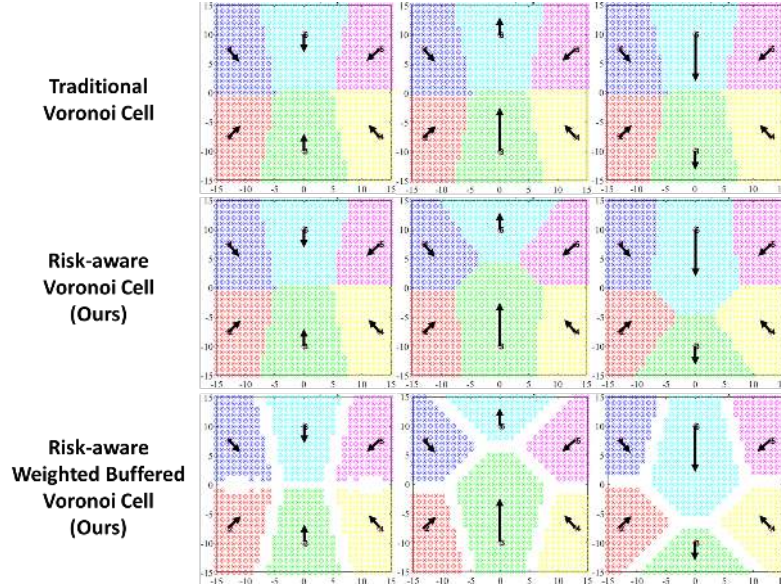


Fig. 1 Our proposed Risk-aware Weighted Buffered Voronoi Cells v.s. traditional Voronoi Cells.

Given their ability to accomplish large-scale tasks, multi-agent systems have been studied in a number of application domains, e.g. search and rescue [1], environmental sampling and exploration [2, 3], and precision agriculture [4]. To keep a multi-agent system safe, a collision-free configuration is required for every pairwise agent interaction in the system. In [5, 6] reciprocal velocity obstacles are used to control agents by assuming symmetric collision-avoidance reasoning of other agents. Safety barrier certificates have been explored to achieve collision-free motion by characterizing the joint admissible control space for the robot team [7, 8]. Methods in [9, 10] use Model

Predictive Control-based methods for individual agent trajectory planning with receding horizons, treating all other agents as dynamic obstacles. However, these methods rely on an individual robot controller to constrain their own motion based on assumed behavior models of the surrounding robots and environments to achieve collision avoidance.

On the other hand, centralized control methods for multi-agent systems can be computationally expensive, and inter-agent communication may not be always available. Therefore, it is favorable to explore decentralized control approaches for multi-agent collision avoidance. Different approaches have been explored to translate centralized control into a decentralized setting, by splitting the constraints and separately solving individual optimization problems with split constraints, so that agents only need to make decisions based on local information, without the need to predict what others are going to do.

As a model-based approach, Control Barrier Functions (CBF) [11] is a powerful tool to ensure robot safety. It is widely applied as a constraint in optimization-based controllers to constrain the robot motion, so that as long as the robot is inside the defined safe set initially, then CBF can always ensure that the robot stays inside the safe set with formally provable guarantees [8, 12]. There have been some efforts in exploring decentralized CBF by splitting the joint admissible control space in different ways. [12] partitions the constraints based on agents' various actuation limits. [13, 14] demonstrate how to divide the constraints given the known information of agent social personalities, egoistic vs. altruistic.

Although decentralized CBF-based methods work well, they require all robots in the multi-agent system to use decentralized CBF-based controllers, and do not generalize well. To accommodate the possibility that agents may use different controllers, Voronoi Cell-based methods are more suitable. Voronoi Cells [15] have been extensively studied by partitioning the joint state space of the multi-agent system into individual cells, representing the admissible state space for each individual agent to avoid the collision. The space partition is conducted based on the spatial distance from a given point in the state space to each robot. However, we argue that agents with different motions should have different-sized cells. In this paper, we aim to build Risk-aware Weighted Buffered Voronoi Cells, in which risk refers to the likelihood of possible collision among agents. The intuition is that simply partitioning the state space based on the point-to-agent proximity, as existing Weighted Buffered Voronoi Cells do, cannot reflect the agent motion information well, which is quite crucial in collision avoidance. In other words, the risk of collision between any pairwise robots should not only depend on how close two agents are, but also on how fast they are moving towards each other. Therefore, our goal is to provide agents that pose higher risk to their neighboring agents with relatively larger cells to reflect the heterogeneity among agents, similar to the idea that a fully-loaded truck with higher speed needs a longer distance to stop.

To evaluate the risk from possible collision, we propose a Control Barrier Function (CBF)-inspired risk measurement that gives an agent situational awareness of the dynamic environment it is in from the CBF perspective. One advantage of our proposed CBF-inspired risk measurement is that the informativeness and expressiveness of CBF

can be leveraged for risk evaluation, which characterizes pairwise agent safety based on various factors including agent positions, motion, and safety radius without requiring agents to actually carry the CBF-based controllers. Different from most existing works in which CBF is merely used as a binary verification of whether the system is still safe given the nominal control, in this work, our proposed CBF-inspired risk measurement characterizes to what extent the system is safe or unsafe, and then it is embedded into the Voronoi Cells partitioning to make the best use of the information CBF provides.

As a variant of traditional Voronoi Cells, Buffered Voronoi Cells have been introduced to account for agent safety radius or uncertainty [13, 15–17] and are widely applied in multi-agent collision avoidance. In Weighted Buffered Voronoi Cells (WBVC), the boundaries of the adjacent cells are biased in a certain way so that the space between two agents is not divided equally. Since our focus is to construct the Weighted Buffered Voronoi Cells in a risk-aware manner, different from other works that only concern safety between pairwise robots, we argue that in order to better characterize an agent’s collision risk, only considering the effect of its neighboring agents is not enough. We also need to consider the neighbors of the neighbors. An example is, for an agent at the center of a crowd being surrounded and another agent at the edge of the crowd, although the pairwise safety requirements are both satisfied, the agent surrounded by the crowd faces higher risk. Therefore, another advantage of our CBF-inspired risk measurement is that it can take the aggregated risk into consideration, so that the cell boundaries are biased towards agents with less relative aggregated risk, leaving them a more constrained state space to proceed with extra caution, with the awareness regarding the existence of its riskier neighbor.

Our **main contributions** are: **1)** We present a CBF-inspired risk measurement to quantify the cumulative risk the agents face in a crowded dynamic environment, especially for multi-agent interaction scenarios; **2)** We propose Risk-aware Weighted Buffered Voronoi Cells (Risk-aware WBVC), which partitions the joint state space of the robot teams and biases the cell boundaries based on CBF-inspired risk measurement, making it generally applicable and extremely suitable for multi-agent decentralized collision avoidance maneuvers; **3)** We provide rigorous proofs on the formal safety guarantee our Risk-aware Weighted Buffered Voronoi tessellation brings, and validate the proposed method in multi-agent collision avoidance scenarios with up to 25 robots.

This article is an extended version of our earlier conference paper, *Decentralized Safe Navigation for Multi-agent Systems via Risk-aware Weighted Buffered Voronoi Cells*, which appeared at AAMAS 2023. In this version, we have expanded the experiment section, in particular we conduct a comprehensive comparison with benchmark algorithms, and provide an in-depth analysis of the strengths of the proposed approach over existing literature.

2 Preliminaries

2.1 Background of Control Barrier Function

Control Barrier Functions (CBF) [11] are used to define an admissible control space for the safety assurance of dynamical systems. One of its important properties is its

forward-invariance guarantee of a desired safety set. Consider a nonlinear system in control affine form: $\dot{x} = f(x) + g(x)u$, where $x \in \mathcal{X} \subset \mathbb{R}^n$ and $u \in \mathcal{U} \subset \mathbb{R}^m$ are the system state and control input with f and g assumed to be locally Lipschitz continuous. A desired safety set $\mathcal{H}$ can be denoted by a safety function $h(x)$: $\mathcal{H} = \{x \in \mathbb{R}^n : h(x) \geq 0\}$. Thus the control barrier function for the system to remain in the safety set can be defined as follows [11]:

Definition 1. (*Control Barrier Function*) Given the aforementioned dynamical system and the set $\mathcal{H}$ with a continuously differentiable function $h : \mathbb{R}^n \rightarrow \mathbb{R}$, then h is a control barrier function (CBF) if there exists a class $\mathcal{K}$ function for all $x \in \mathcal{X}$ such that

$$\sup_{u \in \mathcal{U}} \{\dot{h}(x, u)\} \geq -\kappa(h(x)) \quad (1)$$

We selected the same class $\mathcal{K}$ function $\kappa(h(x)) = \gamma h(x)$ as in [18–20], where $\gamma \in \mathbb{R}^{\geq 0}$ is a CBF design parameter controlling system behaviors near the boundary of $h(x) = 0$. Hence, the admissible control space in (1) can be redefined as $\mathcal{B}(x) = \{u \in \mathcal{U} : \dot{h}(x, u) + \gamma h(x) \geq 0\}$. It is proved in [11] that any controller $u \in \mathcal{B}(x)$ will render the safe state set $\mathcal{H}$ forward-invariant, i.e., if the system starts inside the set $\mathcal{H}$ with $x(t=0) \in \mathcal{H}$, then it implies $x(t) \in \mathcal{H}$ for all $t > 0$ under controller $u \in \mathcal{B}(x)$.

2.2 Weighted Buffered Voronoi Cell

[Summarized from [13]] Consider a bounded, convex environment $Q \subset \mathbb{R}^n$, with individual points $q \in Q$. Within the environment, there are N agents, with the positions of each agent denoted $p_i \in Q$ for $i = \{1, \dots, N\}$. Each agent is assumed to have integrator dynamics,

$$\dot{p}_i = u_i \quad (2)$$

where u_i is the nominal control input for agent i .

The standard Voronoi Partition is defined as:

$$V_i = \{q \in Q \mid \|q - p_i\|^2 \leq \|q - p_j\|^2, \forall j \neq i, \quad i, j \leq N\} \quad (3)$$

where V_i denotes the cell of each agent i . Boundaries between cells are determined based on the Euclidean distance between q and the agents. Therefore, the boundary perpendicularly and equally divides the space between the neighboring agents, leaving them the same distances to the shared boundary.

The Weighted Voronoi Partition is defined as:

$$V_i = \{q \in Q \mid \|q - p_i\|^2 - \omega_i \leq \|q - p_j\|^2 - \omega_j\} \quad (4)$$

where ω_i and ω_j are the weights of each agent, and the Voronoi boundary is moved towards the agent with the larger weight.

The Buffered Voronoi Partition is defined as:

$$V_i = \{q \in Q \mid \|q - p_i\|^2 \leq \|q - p_j\|^2 - \omega_{ij}\} \quad (5)$$

where ω_{ij} defines the cell weightings between agent i and neighbors $j \in \mathcal{N}_i$, creating gaps between agent boundaries as a safety radius buffer. Buffered Voronoi Cells are usually used to guarantee agents will not collide when situated on their boundaries [15, 21].

3 Method

3.1 Control Barrier Function (CBF)-inspired Risk Measurement

Since we are interested in pairwise agent safety, we define the pairwise safety function $h_{ij}(p)$ and safety set $\mathcal{H}$ as:

$$\mathcal{H}(p) = \{p \in \mathcal{P} : h_{ij}(p) = \|p_i - p_j\|^2 - r_{safe}^2 \geq 0, \forall i \neq j\} \quad (6)$$

$p_i, p_j \in \mathbb{R}^2$ for $i, j = \{1, \dots, N\}$ are the positions of any pairwise agents i and j , and $r_{safe} = r_i + r_j$ is the required minimum pairwise safety distance considering both agents' safety radius. The admissible control space for each set of pairwise agents is: $\mathcal{B}_{ij}(p) = \{u \in \mathcal{U} : \dot{h}_{ij}(p, u) \geq -\gamma(h_{ij}(p))\}$. Inspired by CBF, we define our pairwise safety loss function $L_{ij}(p, u) \in \mathbb{R}$ as follows:

$$L_{ij}(p, u) = -\dot{h}_{ij}(p, u) - \gamma h_{ij}(p) - c \quad (7)$$

where c as a constant offset is a very large positive value to ensure $L_{ij}(p, u)$ is always negative to prevent unintended cancel-out when being accumulated later. $u_i, u_j \in \mathbb{R}^2$ are the agents' velocities. The safety loss function $L_{ij}(p, u)$ represents how close the system is to the boundary of the safe set, or how easily a safety violation could occur, under the assumption that both agents move with piecewise-constant velocity [22].

$L_{ij}(p, u)$ describes the risk agent i faces as the potential safety loss when interacting with agent j . For a multi-agent system, the aggregated risk agent i faces posed by surrounding agents $R_i \in \mathbb{R}$ is therefore defined as:

$$R_i = \phi\left(\sum_{j=1}^N L_{ij}(p, u)\right), \quad \forall j \neq i \quad (8)$$

where $\phi(\cdot) \in \mathbb{R} \mapsto \mathbb{R}$ is a mapping function that maps the value of R_i to the range $[0, 1]$. The larger R_i is, the more likely a safety violation is to occur. The proposed risk measurement is simple yet effective: 1) R_i grows with the increased number of agents in the system, as the environment becomes more complex and challenging; 2) R_i varies depending on the changes of states, including positions and motion of other agents as we expected, as it is important to tell how much risk agent i is exposed to even when a collision has not happened yet.

Fig. 2 provides an illustrative example of how risk is calculated for individual agents in a multi-agent interaction scenario. For all six robots, the risk each individual agent faces consists of the pairwise risk generated by the surrounding five agents.

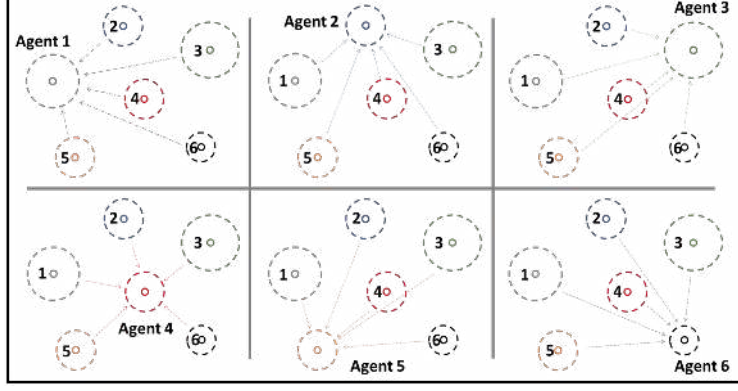


Fig. 2 Propagated risk evaluation for individuals in a multi-agent system. Each arrow represents the risk posed by the pairwise relative movement of the evaluated agent and its neighboring agent.

Note that the proposed CBF-inspired risk measurement does not necessarily require the agents to use CBF-based controllers. We understand that in the real world, agents may use different kinds of controllers, yet it does not prevent them from understanding the risk generated from multi-agent interaction from the CBF perspective, with the mild but reasonable assumption that information about agent states and safety margins are known or observable.

3.2 Risk-aware Weighted Buffered Voronoi Cells (Risk-aware WBVC)

To incorporate the risk measurement information in the Voronoi Cells partition, we extend the Control Barrier Function-inspired risk measurement to any individual point q in the state space being evaluated, by assuming that it is occupied by a static agent with zero velocity. In this way, Eq. 7 and 8 can be generalized from the original agent-to-agent risk measurement to agent-to-point risk measurement.

Definition 2. In a bounded, convex environment $Q \in \mathbb{R}^n$, for any individual point $q \in Q$ and each agent i with $p_i \in Q$ for $i = \{1, \dots, N\}$, the cell V_i of agent i is a **Risk-aware Voronoi Cell**, if

$$V_i = \{q \in Q \mid \|L_{qi}\| \leq \|L_{qj}\|, \forall j \neq i, \quad i, j \leq N\} \quad (9)$$

where $L_{(\cdot)}$ is the pairwise safety loss function calculated in Eq. 7.

Different from traditional Voronoi Cells, we use agent-to-point safety loss function $L_{(\cdot)}$ instead of point-to-point Euclidean Distance, to partition the state space based on the amount of risk agents i and j generate on the individual point q . Therefore, the agent which brings the surrounding environment higher risk has a larger cell.

Definition 3. In a bounded, convex environment $Q \in \mathbb{R}^n$, for any individual point q with $p_q \in Q$ and each agent i with $p_i \in Q$ for $i = \{1, \dots, N\}$, the cell W_i of agent i is

a **Risk-aware Weighted Voronoi Cell**, if

$$\begin{aligned} W_i &= \{q \in Q \mid \|L_{qi}\| - \omega_i \leq \|L_{qj}\| - \omega_j, \quad \forall j \neq i, \quad i, j \leq N\} \\ \omega_i &= \frac{1}{2}(p_i - p_j)^T(u_i + u_j) + \frac{1}{4}(R_i - R_j) \cdot \|(p_i - p_j)^T(u_i - u_j)\| \\ \omega_j &= -\omega_i \end{aligned} \quad (10)$$

where $L_{(\cdot)}$ is the pairwise safety loss function calculated in Eq. 7 and $R_{(\cdot)}$ is the accumulated risk over the individual agents calculated in Eq. 8. $p_{(\cdot)}$ and $u_{(\cdot)}$ are agent position and velocity. $\gamma \in \mathbb{R}$ is the CBF design parameter.

By embedding CBF-inspired accumulated risk measurement into the weight design, the direction of the weight bias is decided based on the relative accumulated risk agent i and j received. Compared to agent j , the larger the relative accumulated risk agent i receives from the rest of the group, the larger weight ω_i it has, and the further the boundary between agent i and j is pushed towards j . To take agent safety radius into account, a buffer is added to the cell boundaries to ensure safety while agents are situated on the boundaries:

Definition 4. In a bounded, convex environment $Q \in \mathbb{R}^n$, for any individual point q with $p_q \in Q$ and each agent i with $p_i \in Q$ for $i = \{1, \dots, N\}$, the cell W_i of agent i is a **Risk-aware Weighted Buffered Voronoi Cell**, if

$$\begin{aligned} W_i &= \{q \in Q \mid \|L_{qi}\| \leq \|L_{qj}\| - \omega_{ij}, \quad \forall j \neq i, \quad i, j \leq N\} \\ \omega_{ij} &= \|r_i + r_j\| \cdot \|\gamma(p_i - p_j) + (u_i - u_j)\| - (p_i - p_j)^T(u_i + u_j) \\ &\quad + [1 - \frac{1}{2}(R_i - R_j)] \cdot \|(p_i - p_j)^T(u_i - u_j)\| \end{aligned} \quad (11)$$

where $L_{(\cdot)}$ is the pairwise safety loss function calculated in Eq. 7 and $R_{(\cdot)}$ is the accumulated risk over the individual agents calculated in Eq. 8. $r_{(\cdot)}$ is agent safety radius, $p_{(\cdot)}$ and $u_{(\cdot)}$ are agent position and velocity. γ is the CBF design parameter.

With the weight design ω_{ij} shown in Eq. 11, a gap between cells with a minimum distance of $(r_i + r_j)$ is added between adjacent cells. Note that when CBF parameter $\gamma = 1$ and $u_i = u_j = 0$, the Risk-aware Weighted Buffered Voronoi Cell and weighting degrade to the traditional Voronoi Cell and Voronoi weighting [15, 21], which considers positional information only.

3.3 Safety Guarantee of Risk-aware WBVC

We propose that a multi-agent system can achieve formally provable safety guarantees in collision avoidance by utilizing the proposed Risk-aware WBVC. In this section, we prove that with Risk-aware Weighted Buffered Voronoi Cells, agents are guaranteed not to collide.

Definition 5. In a multi-agent system of N agents, the configuration of agents is collision-free if the distances $\|p_i - p_j\|$ of all pairwise agents i and j satisfy

$$\|p_i - p_j\| \geq (r_i + r_j), \quad \forall i, j \in \{1, \dots, N\}, i \neq j \quad (12)$$

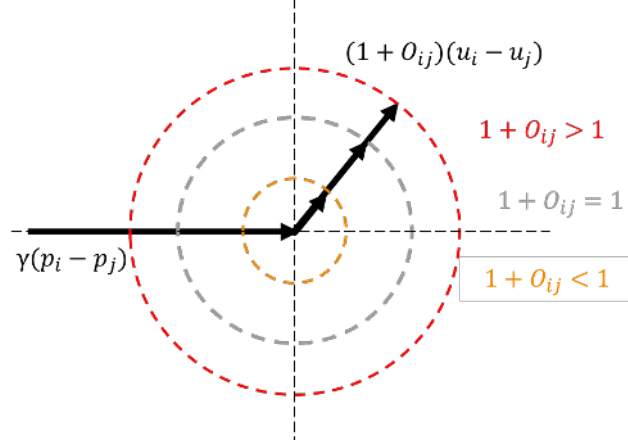


Fig. 3 A geometric illustration of the inequality in Eq. 12.

where p_i, p_j are agent positions and r_i, r_j are agent safety radii.

Now we prove that for any agents in Risk-aware Weighted Buffered Voronoi Cells in collision-free configurations, 1) the cell of each agent is guaranteed to be non-empty; 2) the minimum distance between any two cells is guaranteed to satisfy the collision-free configuration requirement; and 3) the cells of Risk-aware WBVC are non-overlapping.

Lemma 1. (Non-empty Cells). For each agent i with position p_i and cell W_i , $W_i \neq \emptyset$ when the agents are in a collision-free configuration.

Proof. To prove the property of cell non-emptiness, we aim to prove that for any agent i , its position p_i always belongs to W_i . By Def. 4, substitute $q = p_i$ and we have:

$$\begin{aligned} -\gamma(r_i + r_j)^2 &\leq 2(p_j - p_i)^T u_j + \gamma[(p_j - p_i)^2 - (r_i + r_j)^2] - \omega_{ij} \\ \Rightarrow (p_i - p_j)^T [\gamma(p_i - p_j) - 2u_j] - \omega_{ij} &\geq 0 \\ \Rightarrow (p_i - p_j)^T [\gamma(p_i - p_j) + (u_i - u_j) - (u_i + u_j)] - \omega_{ij} &\geq 0 \end{aligned} \quad (13)$$

Substituting ω_{ij} defined in Def. 4 into the equation above, we have

$$\begin{aligned} (p_i - p_j)^T [\gamma(p_i - p_j) + (1 + O_{ij})(u_i - u_j)] \\ \geq (r_i + r_j) \|\gamma(p_i - p_j) + (u_i - u_j)\| \end{aligned} \quad (14)$$

where $O_{ij} = [1 - \frac{1}{2}(R_i - R_j)] \cdot \text{sign}[(p_i - p_j)^T (u_i - u_j)] \in \mathbb{R}$. It is straightforward that Eq. 14 always holds true. A geometric illustration is demonstrated in Fig. 3 for easier understanding. The black arrow on the left denotes the vector $\gamma(p_i - p_j)$, and the black arrows on the right denote the vector $(1 + O_{ij})(u_i - u_j)$ with possibly different values of $(1 + O_{ij})$. The dashed circles represent the regions vector $(1 + O_{ij})(u_i - u_j)$ is in corresponding to different values of $(1 + O_{ij})$. Since $\|p_i - p_j\| \geq (r_i + r_j)$ always stays true with the collision-free configuration, it is illustrative that, if we can prove that the projection magnitude of vector $(\gamma(p_i - p_j) + (1 + O_{ij})(u_i - u_j))$ along the direction of vector $(p_i - p_j)$ is always greater than or equal to that of vector $(\gamma(p_i - p_j) + (u_i - u_j))$

along the direction of vector $(p_i - p_j)$, then the inequality always stays true. Note that the two vectors $(\gamma(p_i - p_j) + (1 + O_{ij})(u_i - u_j))$ and $(\gamma(p_i - p_j) + (u_i - u_j))$ share the same starting point. The ending point of $(\gamma(p_i - p_j) + (u_i - u_j))$ lies on the grey dashed circle, and the ending point of $(\gamma(p_i - p_j) + (1 + O_{ij})(u_i - u_j))$ lies on the red, grey or yellow dashed circle, depending on the value of $(1 + O_{ij})$. Now this problem boils down to the comparison of two projections. As shown in Fig. 3, when the vector $(u_i - u_j)$ is in the half-plane on the right (when $(p_i - p_j)^T(u_i - u_j) > 0$), $O_{ij} = [1 - \frac{1}{2}(R_i - R_j)] \cdot \text{sign}[(p_i - p_j)^T(u_i - u_j)] > 1$, the ending point of $(\gamma(p_i - p_j) + (1 + O_{ij})(u_i - u_j))$ lies in the right half-plane of the red dashed circle, and the projection magnitude of $(\gamma(p_i - p_j) + (1 + O_{ij})(u_i - u_j))$ is greater than that of $(\gamma(p_i - p_j) + (u_i - u_j))$. This statement also holds true when the vector $(u_i - u_j)$ is in the half-plane on the left (when $(p_i - p_j)^T(u_i - u_j) < 0$), $O_{ij} = [1 - \frac{1}{2}(R_i - R_j)] \cdot \text{sign}[(p_i - p_j)^T(u_i - u_j)] < 0$, meaning the ending point of $(\gamma(p_i - p_j) + (1 + O_{ij})(u_i - u_j))$ lies in the left half-plane of the yellow dashed circle. If the vector $(u_i - u_j)$ sits on the vertical half-planes' boundary, the two projections' magnitude is the same. Thus, it is proved that no matter where the vector $(u_i - u_j)$ lies, it is always guaranteed that Eq. 14 stays true. Therefore, the cells in Risk-aware WBVC $W_i \neq \emptyset$. $\square$

Lemma 2. (*Minimum Distance between Cells*). For agent i and j with Cells W_i and W_j , any two random points $\bar{p}_i \in W_i$ and $\bar{p}_j \in W_j$, $\|\bar{p}_i - \bar{p}_j\| \geq (r_i + r_j)$.

Proof. Given Def. 4, considering $q = \bar{p}_i$ and $\bar{p}_j$ respectively, we have

$$\begin{aligned} & \| -2(p_i - \bar{p}_i)^T(u_i - \bar{u}_i) - \gamma((p_i - \bar{p}_i)^2 - (r_i + r_j)^2) - c \| \leq \\ & \| -2(p_j - \bar{p}_i)^T(u_j - \bar{u}_i) - \gamma((p_j - \bar{p}_i)^2 - (r_i + r_j)^2) - c \| - \omega_{ij} \\ & \| -2(p_j - \bar{p}_j)^T(u_j - \bar{u}_j) - \gamma((p_j - \bar{p}_j)^2 - (r_i + r_j)^2) - c \| \leq \\ & \| -2(p_i - \bar{p}_j)^T(u_i - \bar{u}_j) - \gamma((p_i - \bar{p}_j)^2 - (r_i + r_j)^2) - c \| - \omega_{ji} \end{aligned} \quad (15)$$

By adding the two equations together, we have

$$-2(\bar{p}_i - \bar{p}_j)^T[\gamma(p_i - p_j) + (u_i - u_j)] \leq -(\omega_{ij} + \omega_{ji}) \quad (16)$$

The sum of the two weightings is

$$\omega_{ij} + \omega_{ji} = 2\|r_i + r_j\| \cdot \|\gamma(p_i - p_j) + (u_i - u_j)\| + 2\|(p_i - p_j)^T(u_i - u_j)\| \quad (17)$$

Substituting this expression into Eq. 16, this reduces to

$$\begin{aligned} & (\bar{p}_i - \bar{p}_j)^T[\gamma(p_i - p_j) + (u_i - u_j)] \\ & \geq \|r_i + r_j\| \cdot \|\gamma(p_i - p_j) + (u_i - u_j)\| + \|(p_i - p_j)^T(u_i - u_j)\| \end{aligned} \quad (18)$$

For two vectors a and b , it is known that $\|a\|\|b\| > \|a^T b\|$. Let $a = (\bar{p}_i - \bar{p}_j)$ and $b = [\gamma(p_i - p_j) + (u_i + u_j)]$; we can then state

$$\begin{aligned}
\|\bar{p}_i - \bar{p}_j\| &\geq \frac{(\bar{p}_i - \bar{p}_j)^T [\gamma(p_i - p_j) + (u_i - u_j)]}{\|\gamma(p_i - p_j) + (u_i - u_j)\|} \\
&\geq \frac{\|r_i + r_j\| \cdot \|\gamma(p_i - p_j) + (u_i - u_j)\| + \|(p_i - p_j)^T (u_i - u_j)\|}{\|\gamma(p_i - p_j) + (u_i - u_j)\|} \\
&= (r_i + r_j) \left(1 + \frac{\|(p_i - p_j)^T (u_i - u_j)\|}{\|\gamma(p_i - p_j) + (u_i - u_j)\|}\right) \\
&\geq \|r_i + r_j\|
\end{aligned} \tag{19}$$

thus proving that $\|\bar{p}_i - \bar{p}_j\| \geq (r_i + r_j)$ for any two points $\bar{p}_i \in V_i$ and $\bar{p}_j \in V_j$, for all $i \neq j$. $\square$

Lemma 3. (*Non-overlapping Cells*). For a point $\bar{p}_i$ belonging to W_i , then $\bar{p}_i \notin W_j$ for all $j \neq i$, and $W_i \cap W_j = \emptyset$.

Proof. For a point $q \in W_i$, it must satisfy the cell definition in Def. 4:

$$\|L_{qi}\| \leq \|L_{qj}\| - \omega_{ij} \tag{20}$$

Assuming $q \in W_j$ is true at the same time, by definition we have

$$\begin{aligned}
\|L_{qj}\| &\leq \|L_{qi}\| - \omega_{ji} \\
&\leq \|L_{qj}\| - \omega_{ij} - \omega_{ji} \\
0 &\leq -2\|r_i + r_j\| \cdot \|\gamma(p_i - p_j) + (u_i - u_j)\| - 2\|(p_i - p_j)^T (u_i - u_j)\|
\end{aligned} \tag{21}$$

From this contradiction, it is proved that $q \notin W_j$. Since any point within a cell cannot belong to any other cell at the same time, we conclude that $W_i \cap W_j = \emptyset, \forall i \neq j$. $\square$

3.4 Collision-free Navigation with Risk-aware WBVC

We have proved in the previous section that, with our proposed Risk-aware WBVC, being collision-free is ensured when agents are within their own cells. With our three proven Risk-aware WBVC properties, it is straightforward and has been proved in [13] (Theorem 1) that, for agents navigating towards target positions which are located outside the cells, projecting the goal positions inside the cells can always guarantee collision-free navigation performance.

Now we present our algorithm for multi-agent safe navigation utilizing the proposed Risk-aware Weighted Buffered Voronoi Cells in Algorithm 1.

where $\bar{g}_i$ is the desired goal position of agent i , and $\hat{g}_i$ is the projected goal position if the goal is outside the cell W_i . g_i^* is the goal position fed into a move-to-goal controller or any other kind of controller that drives the robot to move towards the goal. With our proposed algorithm, while the goal has not been reached, the agent will

Algorithm 1 Safe Navigation in Risk-aware WBVC

Require: Collision-free $p(t_0), u(t_0), \bar{g}, \forall i \in \{1, \dots, N\}$

```
while  $\|\bar{g}_i - p_i\| > 0, \forall i \in \{1, \dots, N\}$  do
    Calculate pairwise safety loss function  $L_{ij}$  (Eq. 7)
    Calculate the accumulated risk  $R_i$  (Eq. 8)
    Update weights  $\omega_{ij}$  (Def. 4)
    Update Voronoi Cell Tessellation  $W_i$  (Def. 4)
    if  $\bar{g}_i \in W_i$  then
        Set  $g_i^* = \bar{g}_i$ 
    else
        Find intersection  $\hat{g}_i$  of the cell boundary and the connecting line between  $p_i$ 
        and  $\bar{g}_i$ 
        Set  $g_i^* = \hat{g}_i$ 
    end if
    Update control policy  $\dot{p}_i = -k(g_i^* - p_i)$ 
end while
```

first calculate L_{ij} and R_i with our CBF-inspired risk measurement. By updating the Risk-aware Weighted Buffered Voronoi Cells, the heterogeneity among agents in terms of different levels of risk exposure and risk generation ability is reflected in different sizes and shapes of the cells.

4 Simulation & Discussion

In this section, we demonstrate the validity and effectiveness of the proposed method in multi-agent swapping games. The goal is for pairwise agents to switch positions with each other without any collision. We set the agent safety radius as $0.2m$ and the right-hand heuristic is applied for deadlock resolution [23, 24]. We show that our approach could be applied to agents with unicycle dynamics and more complicated differential-drive two-wheeled mobile robots.

4.1 Geometric Illustration of Risk-Aware WBVC

To elucidate the distinction between our proposed tessellation and the traditional Voronoi tessellation, we first offer a geometric illustration of both approaches in a static configuration with six agents. In this example, the agents have unicycle dynamics, and we employ a nonlinear inversion method [25] to map the desired velocity to the unicycle dynamics of mobile robots without compromising the safety guarantee [26].

The comparison is shown in Fig. 1. The three rows correspond to traditional Voronoi Cells, Risk-aware Voronoi Cells and Risk-aware Weighted Buffered Voronoi Cells. The cells of the six agents are marked in different colors and the black arrows indicate the velocity of the agents with different magnitudes and directions. The agents in all plots share the same positions, but with different velocity settings corresponding to the three columns.

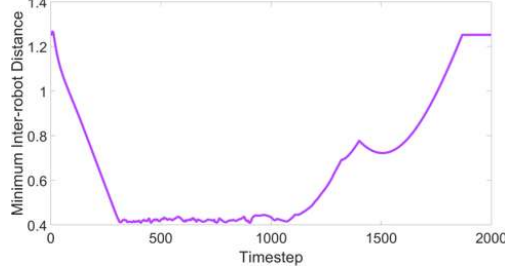


Fig. 4 The minimum inter-robot distance among 6 robots.

In the first row, we can see that since the tessellation is conducted solely based on positional information and does not take agent motion into account, no matter where the agents are heading and how fast they are moving, the cells are equally partitioned. In the second row, we use the proposed CBF-inspired risk measurement to take not only agent position but also motion into account for tessellation, without any weight or buffer. We can see that the sizes and shapes of the cells change when the agent takes different actions. Since we are not partitioning the space based on the point-to-agent euclidean distance, the shared cell boundaries are no longer perpendicular to the line between neighboring agents. This makes sense, since if we take agent motion into consideration, the generated risk is higher in the direction it moves than in the opposite direction, and it is therefore unnecessary to leave the same large cell space behind the agent. Our proposed Risk-aware WBVC is presented in the third row, and we observe that the lower the risk the agent poses to its surrounding environment compared to its neighboring agents, the smaller its cell is, warning it of the presence of its dangerous neighbor. The higher the risk the agent is exposed to, the larger cell it is given to grant it additional space for safe maneuvers.

Another effective example of our Risk-aware WBVC is shown in Fig. 10, in which the positions of all agents are fixed and the agent velocities are randomized. From the 10 randomized trials, it is shown that our proposed Risk-aware WBVC can better reflect the risk quantified via the CBF-based risk measurement, compared to the traditional Weighted Buffered Voronoi Cells that will have the same and equal cells.

4.2 Validity of Risk-Aware WBVC

Next, we demonstrate the validity of Risk-aware WBVC by verifying its safety assurance. We first show the six-agent position swapping game in Fig. 11. The numbers in black are agent indices and the numbers in red are target position indices, indicating to which agent they belong. We observe that the agents are able to dynamically compute and update their Voronoi Cells in a risk-aware manner. The inter-robot distance over time is recorded in Fig. 4. The y-axis represents the minimum inter-agent distance. Since we set the safety radius of individual agents to be $0.2m$, the minimum inter-agent distance should be larger than or equal to $0.4m$. The plot shows that the safety requirement is satisfied in the six-agent position-swapping game.

In the second example, we consider a more complicated and challenging environment with 16 robots in total. We ran 3 trials with randomly assigned pairwise

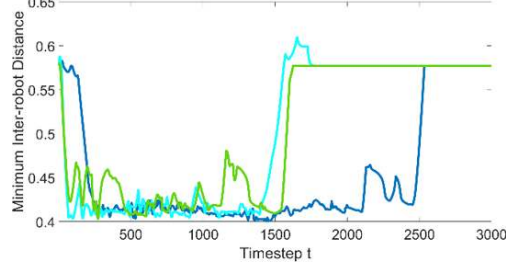


Fig. 5 The minimum inter-robot distance among 16 robots over 3 trials.

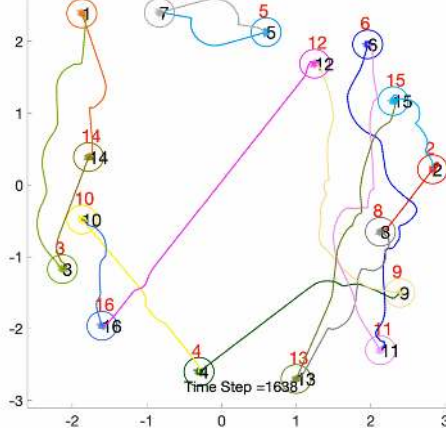


Fig. 6 Trajectories of sixteen robots.

counterparts for agents to swap positions. We demonstrate the swapping process of one trial in Fig. 12. For better visualization, we separately show agent trajectories in Fig. 6. The minimum inter-robot distance over time of the 3 trials is shown in Fig. 5, with the green line corresponding to the demo case in Fig. 12. It is observed that no safety violation happens during the whole process and the agents are able to safely navigate to their target positions. The risk evaluation is conducted and the risk measurement is leveraged to update the Weighted Buffered Voronoi Cells in a risk-aware manner.

4.3 Comprehensive Benchmark Comparison

To further analyze the performance of our risk-aware WBVC, we select a) Weighted Buffered Voronoi Cells (WBVC) and b) Truncated Hybrid Reciprocal Velocity Obstacles (THRVO) as our benchmark algorithms. we assess algorithm performance across varying numbers of robots in the scene, reflecting the evolution of dense environments and escalating interaction complexity. In this evaluation, the robots are implemented with differential drive kinematics with maximum allowable linear and angular speeds to represent more realistic robot dynamic systems. The comparison is still conducted in a robot position-swapping game with robots swapping start and end positions between one another.

4.3.1 Benchmark Algorithms

- a) Weighted Buffered Voronoi Cells (WBVC): This is the non-risk-aware version of our proposed approach, with details described in Section 2.2.
- b) Truncated Hybrid Reciprocal Velocity Obstacles (THRVO) [27]: This approach extends the fundamental concept of velocity obstacles (VO) in navigation, as initially proposed by Fiorini et al. [28], to enable collision avoidance in environments with multiple dynamic obstacles. VO operate by establishing a geometric representation of the collective positions and velocities of the agent and each obstacle. This representation is then utilized to ascertain a range of permissible collision-free velocities for the agent. This enables the agent to navigate safely by selecting velocities within this permissible range. Later on, reciprocal velocity obstacle (RVO) [29] and Hybrid reciprocal velocity obstacles (HRVO) [30, 31] were proposed to further incorporate the anticipated action of others and directional heuristics that guide robots in reaching consensus on passing sides into the controller design, thereby mitigating "reciprocal dances". Instead of considering the risk of collision in all future timesteps, THRVO [27] only assesses possible collisions in finite time and therefore can effectively avoid deadlock situations without compromising immediate safety. THRVO is chosen as the second baseline approach as it has been widely used for benchmark comparison in the existing literature of decentralized multi-agent navigation [32–36], and we follow [37] for detailed implementation.

Comprehensive Evaluation on Safety and Efficiency of Risk-Aware WBVC

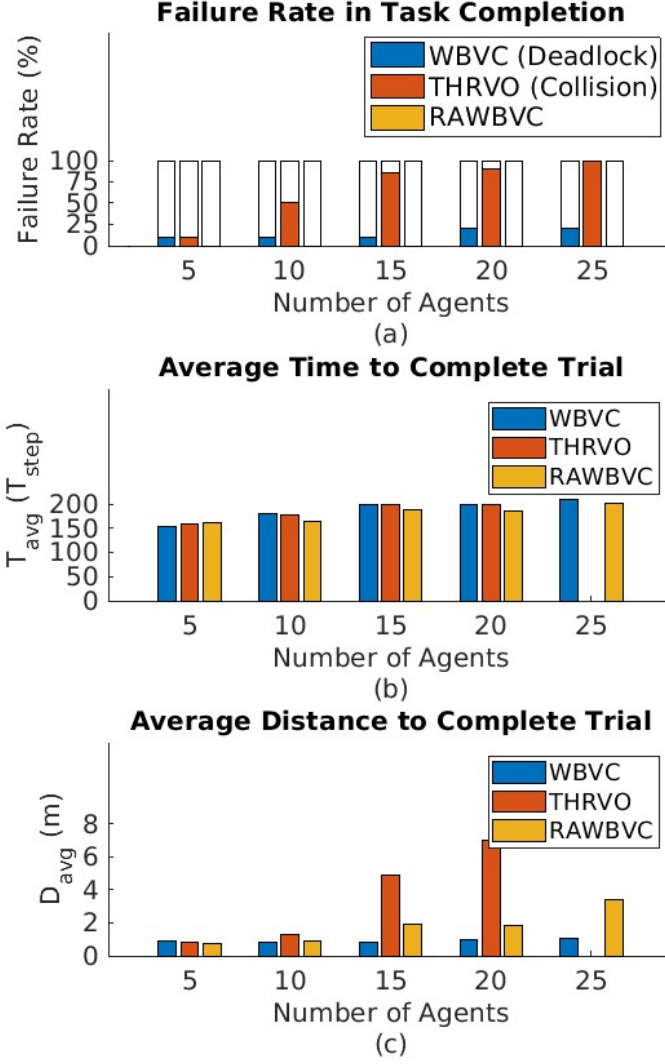


Fig. 7 A comparison of safety and performance across multiple navigation approaches for a robot switching game with varying numbers of agents.

4.3.2 Analysis on Safety

To compare their performance on safety, we conduct 100 trials with randomized initial conditions (including robot positions and velocities) for each batch of robots in

different numbers. The failure rate in task completion for every batch was recorded in Fig. 7(a). Two kinds of failure are considered: 1) inter-robot collision happens, and 2) the task is not completed within the required time due to deadlock situations. To analyze safety performance, we only focus on the collision rate here. It is observed that no collision happens for both WBVC and Risk-Aware WBVC. However, with the increasing number of robots in the batch, the collision rate increases for THRVO.

To better understand how the safety violation happened for THRVO, we analyze one of the failure scenarios where THRVO encounters a collision, contrasting it with our method. In Fig. 8, we track the minimum inter-agent distance over time. The solid red and green lines depict the average minimum distance for THRVO and our approach respectively, while the shaded areas denote the variance across all agent pairs. The safety margin is indicated by a black dashed line. It’s crucial for the shaded areas to consistently remain above this line to ensure the system’s safety. Here we call special attention to the moment when the red region falls below the safety margin, corresponding with a collision between two agents, as highlighted in the zoomed-in segment of the figure.

One primary factor contributing to the high collision rates with THRVO is its reliance on velocity obstacle (VO) based approaches. These methods construct a geometric representation of an agent’s permissible velocities, considering all potential collision scenarios with other agents treated as dynamic obstacles maintaining their observed velocities. Essentially, an agent’s allowable velocity space is determined solely by the velocities of others. Additionally, VO-based approaches assume instantaneous acceleration or deceleration in all directions for every agent, presuming that each will maintain its current velocity in the next time step. However, this assumption becomes problematic, especially when agents don’t adhere to single integrator dynamics. For instance, in our simulation utilizing differential drive agent dynamics, the assumption of instantaneous acceleration or deceleration doesn’t hold, leading to collisions. Conversely, the motion information considered by our Risk-aware WBVC is not limited to velocities, but can also include acceleration, or other higher-order control inputs (using High-Order CBF) like roll, pitch, yaw, thanks to CBF’s strong descriptiveness and generality. This comparison also demonstrates the seamless integration of our proposed Risk-Aware Weighted Buffered Voronoi Cells (WBVC) with various robot dynamics.

4.3.3 Analysis on Efficiency

Next, we evaluate the task efficiency of the three approaches from the following three perspectives, the deadlock rate, the average time to complete trials, and the average distance to complete trials.

The deadlock rate for different batches of robots is reported in Fig. 7a, marked in blue. It is observed that among the three approaches we compared, the only one that encountered deadlock situations is WBVC. The deadlock rate generally increases with the number of robots in the environment, meaning the more crowded the navigation scenario is, the more challenging for WBVC to avoid deadlock.

The only difference between WBVC and our proposed Risk-aware WBVC is whether the risk information is taken into account. This is important for multi-robot

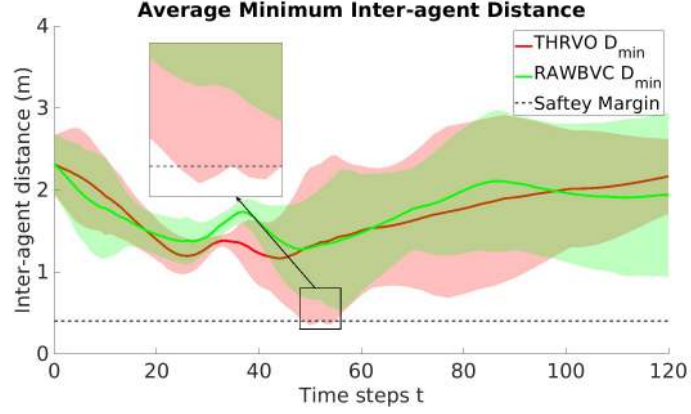


Fig. 8 The average minimum inter-robot distances in a simulation among 10 robots in a robot switching game using multiple control approaches.

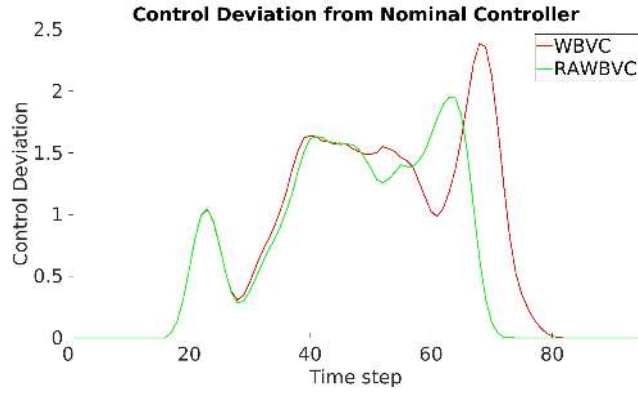


Fig. 9 The control deviation of a single robot of multiple methods from a nominal controller (straight to goal) in a robot switching game.

navigation in dense environments, as with the proposed CBF-inspired risk measurement, our Risk-aware WBVC can reason over the indirect impact from the neighbors of neighbors. Our approach captures the risk posed by the agents which are not immediate neighbors to the ego agent but may still be troublesome by impacting the ego's neighbors. A simple example would be, if we consider two agents A and B, assuming another agent C is on the other side of B, the risk A receives should not remain the same as the case without C's presence. Although C will not pose an immediate collision threat to A, its presence reduces the admissible space of B, bringing an additional constraint for B to stay collision-free, and therefore indirectly makes it more challenging for A to stay safe. This ability to take the indirect influence of the non-immediate neighbors into account differs our approach from the two benchmark baselines.

For better illustration, we provide an example scenario of the deadlock situation that WBVC got in in Fig. 9.

[new edits start here](#)

The results presented subplot (a) of Fig. 7 shows the percentage of completed trials for each approach. Subplots (b) and (c) show the mean performance for each approach across multiple trials. We note that for the simulation with 25 agents no data on the time required to complete a trial, or the distance traveled is provided for the velocity obstacle based approach as all robots failed to reach the goal without colliding. This occurred because the 25-robot scenario was intentionally designed to create high congestion, where the dynamics of robots significantly influence navigation performance. Velocity obstacles assure collision-free navigation only for agents with instantaneous motion dynamics so that the robots can always travel at a safe velocity. Given the complexity of the densely crowded scenario and the differential drive dynamics of the robots, this method failed to complete any trials successfully under these conditions.. We call attention to subfigure (c) which demonstrates the success rate of each approach in simulation. We note that our risk-aware cells were the only approach that was able to successfully navigate all robot switching scenarios by avoiding both collisions and deadlocks. This highlights that our proposed method is a statistically safer performance of our approach compared to other methods, as it does not collide with other robots even when velocity obstacle based approaches would incur collisions. It additionally does not incur deadlocks under normal navigation conditions which the traditional voronoi cells based approach suffers from. Our method results in more consistent navigation especially in more crowded scenarios, since the aggregated risk individual agents receive from others, provides an effective approach to avoid potential deadlocks and crashes.

4.4 Efficiency of Risk-aware WBVC

To evaluate the performance of our proposed risk-aware cells we compare them against other approaches in additional simulations with varying numbers of robots as done in our safety comparison. We again compare Weighted Buffered Voronoi Cells, Truncated Hybrid Reciprocal Velocity Obstacles, and our approach of Risk-Aware Weighted Buffered Voronoi Cells. In these additional simulation the robots are again implemented with differential drive kinematics. We implement a robot position swapping game with robots swapping start and end positions between one another.

In addition to providing insights into safety we provide Figure 7 as a quantitative means of comparing performance metrics across different multi-agent navigation approaches. The results presented in this table represent average performance across multiple trials for each method, and show various trends in performance. It can be noted from (a) of Figure 7 that all methods perform similarly in the total time taken to complete a trial, with Risk-Aware cells generally completing trials slightly faster than traditional cells, and Velocity Obstacles. As seen in (b) of Figure 7 risk-aware cells travel slightly further than the traditional voronoi cells, however, this is an expected result since the boundaries between cells are effected by velocity as well as position in our approach, which forces robots to navigate around one another with a slightly wider safety margin when computing collision free trajectories. The risk-aware cells travel significantly shorter distances than the velocity obstacles at higher numbers of agents showing our approaches ability to scale to larger simulations.

Despite the robots tending to maintain further distances from one another with our proposed risk-aware approach compared to velocity obstacles, the average time for all robots to switch positions tends to be slightly faster as seen in Figure 7 (a) and the total distance traveled by agents is also shorter as seen in Figure 7 (b). The faster completion times for trials occurs primarily because as agents get too close to one another in the velocity obstacles approach it can force certain agents to drastically decrease their overall velocity in order to avoid collisions as there are no safe paths that allow the agents to move at their maximum velocities. The risk-aware cells allow agents to more consistently move at higher speeds by maintaining greater overall separation between agents throughout these trials leading to an overall faster completion time.

Meanwhile, in (c) of Figure 7 we see that traditional WBVC fall into deadlocks more easily than our risk-aware approach, which leads to their longer completion times despite their shorter total distance traveled, and fewer successful completed trials. Compared to non-risk-aware WBVC, our method results in more efficient behavior especially in highly crowded scenarios, since the heterogeneity among agents, in terms of the aggregated risk individuals receive from others, is well reflected in the adaptive Voronoi Cells partition. Meanwhile, non-risk-aware WBVC fall into deadlocks more easily and take longer to resolve them, since it is unable to capture and reflect this kind of heterogeneity among robots.

To better illustrate how our proposed risk-aware approach can more effectively avoid deadlock conditions we present Figure 9 which shows the control deviation from a nominal controller (straight to goal) in a simulation of 10 robots in a highly crowded robot position swapping game. We place both robots using a traditional WBVC and risk-aware WBVC controller approaches in identical starting configurations, and allow them to navigate the robots to the goals.

It can be seen from the plot that for approximately the first 40 time steps from both methods perform similarly, which comes from the fact that the robots start the trial far apart from one another. While the robots are separated by large distances in the simulation risk-aware WBVC perform similarly to traditional WBVC as the accumulated risk from all other agents is low when all robots are distant. However, as the robots approach each other and the area become more congested around $t = 40$ it can be seen that the risk-aware WBVC control begins to diverge from the traditional cells. The risk-aware WBVC have an earlier large deviation around $t = 60$ from the nominal controller, which leads all robots to travel further from one another to avoid centralizing in areas where deadlock could occur. This early intervention from the aggregated risk by our approach allows the risk-aware WBVC to complete the trial more quickly, as the robots do not need to escape a near deadlock scenario that occurs in the traditional WBVC around $t = 70$ due to their late intervention.

4.5 Tunable Overall Risk Sensitivity

Compared to traditional Voronoi Cells that only compute cells based on positional information, by quantifying risk from a CBF perspective, our proposed Risk-aware WBVC also characterizes various levels of the overall risk sensitivity of the multi-agent system.

It is demonstrated in Fig. 13 that for a given set of agent positions and velocities, by tuning the CBF design parameter γ , the Risk-aware WBVC is computed with different levels of risk tolerance. The smaller γ is, the more conservative the multi-agent system behaves, and the larger γ is, the more aggressive the overall system is. For higher overall risk sensitivity, extra cautions are taken by the multi-agent system by adaptively enlarging gaps in between the cells for enhanced safety performance.

Note that different levels of system aggressiveness or conservativeness in our Risk-aware WBVC are not equivalent to simply using larger or smaller buffers in traditional Weighted Buffered Voronoi Cells (WBVC). In our Risk-aware WBVC, the cells' shapes and sizes are also different, characterizing the possible heterogeneity of the overall risk sensitivity. Mathematically γ serves as a balancing parameter between the influence of the positional information and the motion information in cell partitioning, and the larger γ is, the closer Risk-aware WBVC is to traditional WBVC. As mentioned previously, our Risk-aware WBVC is a generalized version of traditional WBVC, and it degrades to traditional WBVC when $\gamma = 1$ and the agent motion information is set to zero when computing cells, which is also verified in the proof of Lemma 2 that our Risk-aware WBVC provides a tight lower bound on the minimum distance between cells compared to that of traditional WBVC.

5 Conclusion

In this paper, we propose the Risk-aware Weighted Buffered Voronoi Cells (Risk-aware WBVC), which utilizes novel Control Barrier Function-inspired risk measurement to evaluate the risk each individual agent faces and the accumulated risk from inter-agent interaction. Rigorous proofs are provided for formally provable guarantees for multi-agent collision-free navigation with our Risk-aware WBVC. The effectiveness and safety performance of the proposed approach is demonstrated in multi-agent position-swapping games with up to sixteen robots. For future work, we aim to extend Risk-aware WBVC for safe navigation under uncertainty.

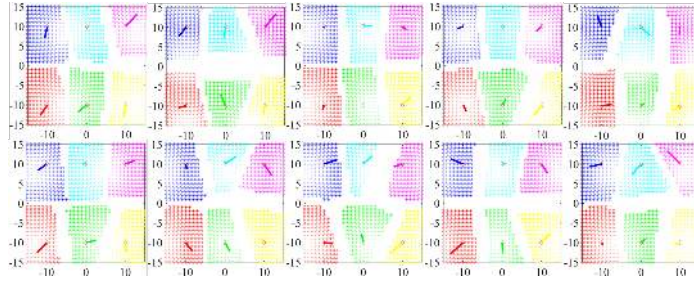


Fig. 10 Risk-aware WBVC with randomized agent velocities at the same positions. The colorful lines represent the randomized agent velocities with different magnitudes and directions.

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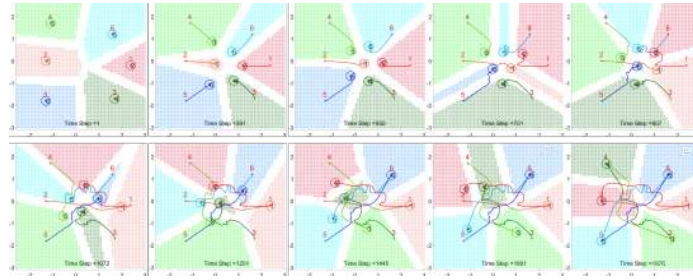


Fig. 11 Our proposed Risk-aware Weighted Buffered Voronoi Cell in a 6-robot position swapping game.

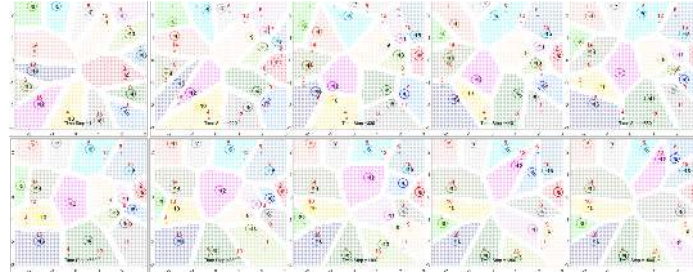


Fig. 12 Our proposed Risk-aware Weighted Buffered Voronoi Cell in a 16-robot position swapping game.

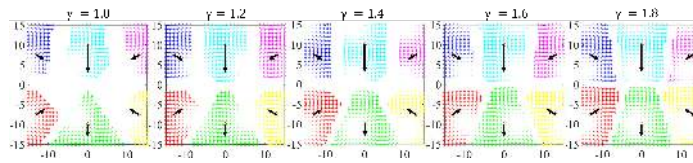


Fig. 13 Different levels of multi-agent system's sensitivity to risk. The smaller γ is, the more sensitive the multi-agent system is to risk and therefore extra cautions are taken when computing Risk-aware WBVC.

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